



BLG640E Security in Financial Information Systems

Syllabus

Spring 2026

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Motivation

Sequoia Capital is hacked due to phishing scam



Sequoia Capital, one of the largest and most successful venture capital firms in the world, has told its investors that some of their personal and financial information may have been accessed by a third party....

<https://www.securitymagazine.com/articles/94652-sequoia-capital-is-hacked-due-to-phishing-scam>

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Motivation

Top 5 Cyber Threats Facing Banks in 2020



With all the cyber threats that exist today, banks are more vulnerable than ever to becoming the next victim of a malicious cyberattack.....

<https://hubsecurity.io/top-5-cyber-threats-facing-banks/>

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Motivation

Report reveals how many financial institutions have experienced a cyberattack



Nearly 70% of financial institutions globally have experienced a cyberattack, and nearly half reported experiencing an attack within the last 12 months.....

<https://www.insurancebusinessmag.com/us/news/cyber/report-reveals-how-many-financial-institutions-have-experienced-a-cyberattack-222285.aspx>

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Outline

Week/Date	Content
1 / 10.02.2026	Course overview and requirements
2 / 17.02.2026	Financial systems and information security (HW and project announcement)
3 / 24.02.2026	Mathematical background for applied cryptography
4 / 03.03.2026	Applied cryptography – 1
5 / 10.03.2026	Applied cryptography – 2
17.03.2026	SPRING BREAK
6 / 24.03.2026	Trust models and identity management
7 / 31.03.2026	Authentication
8 / 07.04.2026	Access control and multilevel security modelling (Project submission deadline : 12.04.2026)
9 / 14.04.2026	Financial markets and currency
10 / 21.04.2026	Blockchain – 1
11 / 28.04.2026	Blockchain – 2
12 / 05.05.2026	MIDTERM
13 / 12.05.2026	Emerging topics or Project presentations (tentative)
14 / 19.05.2026	Project presentations

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Grading

Grading Policy

- Homework : 20% (paper presentation assigned by instructor)
- Term Project : 40% (%30 report + %10 presentations) (will be a research project)
- Midterm : 40%
- Extra Points : ?

Text Book: No mandatory textbook!

Other References:

- William Stallings, *"Cryptography and Network Security: Principles and Practice"*, 7th edition, 2017.
- I. Bashir, *"Mastering Blockchain"*, Packt Publishing Ltd., Livery Place, 35 Livery Street, Birmingham, B32PB, UK, 2017
- G. Karame and E. Androulaki, *"Bitcoin and Blockchain Security"*, ARTECH HOUSE, 685 Canton, Street, Norwood, MA 02062, 2016.
- J. Keyes, *"Financial Services Information Systems"*, Auerbach Publications, CRC Press Taylor & Francis, FL, 2000

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